

BSCS-VII (GE-261)
TECHNICAL & BUSINESS WRITING

1. Overview of Technical Reporting

Technical reporting is the process of documenting and communicating complex information in a clear and structured way. It is a fundamental skill for professionals in science, technology, and engineering, as it allows them to share findings, explain processes, and propose solutions effectively. The primary goal is to inform the reader and provide them with the necessary data to make a decision or take action.

Key Characteristics:

- **Clarity:** Use simple, direct language. Avoid jargon where possible, or if it's necessary, define it clearly.
- **Accuracy:** All facts, data, and figures must be precise and verifiable.
- **Objectivity:** Present information in an unbiased manner. Stick to the facts and avoid personal opinions.
- **Structure:** Reports follow a logical, organized format (e.g., introduction, body, conclusion, and appendices) to help readers navigate the information easily.
- **Audience-focused:** Tailor the content and language to the specific knowledge level of the intended readers.

Technical reports serve a variety of purposes, such as summarizing research, documenting a project's progress, proposing new initiatives, or detailing the results of an experiment. They are essential for professional communication and accountability.

2. Use of Library and Information Gathering

Gathering reliable information is the first and most critical step in creating a technical report. A library, both physical and digital, is a primary resource for this. Using it effectively ensures your information is credible and accurate.

Steps for Information Gathering:

1. **Define Your Topic:** Clearly identify the subject of your report. This helps you narrow your search and stay focused.
2. **Identify Keywords:** List keywords and search terms related to your topic. Think of synonyms and related concepts to broaden your search.
3. **Choose Your Sources:**

- **Primary Sources:** Original research, raw data, interviews, and lab reports. These are firsthand accounts.
- **Secondary Sources:** Interpretations or analyses of primary sources, such as textbooks, articles, and literature reviews.
- **Tertiary Sources:** Compilations of primary and secondary sources, such as encyclopedias and dictionaries.

4. **Use Library Tools:**

- **Online Databases:** Access academic journals (e.g., IEEE Xplore, ACM Digital Library), articles, and research papers.
- **Library Catalog:** Search for books, e-books, and physical materials.
- **Librarian Assistance:** Librarians are experts in research and can help you find specialized resources.

5. **Evaluate Sources:** Not all information is created equal. Check the credibility of your sources by considering the author's expertise, the publication date, and the reputation of the publisher. A reliable source is typically peer-reviewed and published by a reputable academic institution or organization.

3. **Administering Questionnaires**

Questionnaires are a vital tool for gathering **primary data** directly from people. They are used to collect information on opinions, behaviors, and preferences, and are particularly useful for gathering quantitative data (numbers) that can be analyzed statistically.

Key Steps for Creating and Administering a Questionnaire:

1. **Define the Objective:** What specific information do you need to collect? Your questions should directly relate to your research goals.
2. **Design the Questions:**
 - **Keep it Simple:** Use clear, unambiguous language.
 - **Avoid Leading Questions:** Do not phrase questions in a way that suggests a desired answer.

Use Question Types:

- **Open-ended:** Allow for detailed answers (e.g., "What are your thoughts on...?").

- **Closed-ended:** Provide specific answer choices (e.g., Yes/No, multiple choice, rating scales). These are easier to analyze.

3. **Structure the Questionnaire:**

- Start with a brief introduction explaining the purpose of the survey and ensuring confidentiality.
 - Group related questions together.
 - End with a "Thank you" message.
4. **Pilot Test:** Test the questionnaire on a small group before the main distribution. This helps you identify confusing questions or errors.
5. **Distribute and Collect:** Choose a method for distribution, such as online survey tools (Google Forms, Survey Monkey), email, or in-person. Collect the responses and organize the data for analysis.

4. **Reviewing the Gathered Information**

Once you have collected your information—whether from a library, questionnaires, or other sources—you must carefully review and analyze it. This step turns raw data into meaningful insights for your report.

Process of Reviewing Information:

1. **Organize Your Data:** Create a system for organizing your notes, survey responses, and source materials. This can be as simple as using folders, a spreadsheet, or specialized note-taking software.
2. **Evaluate for Relevance and Credibility:** Go through each piece of information and ask:
 - Is this relevant to my report's purpose?
 - Is the source credible and reliable?
 - Is the information up-to-date?
3. **Synthesize and Summarize:** Look for connections and patterns across your different sources. Synthesize the information by combining ideas from various sources and summarizing the key findings.
4. **Identify Gaps:** During the review, you may discover that some questions remain unanswered. This is an opportunity to conduct further research to fill in the missing information.

5. **Critique and Analyze:** Don't just accept the information as is. Critically analyze it. For example, if you have two sources that contradict each other, note the discrepancy and consider why that might be. This critical thinking is what makes a technical report insightful and valuable.